PW SKILLS

Advanced Data Visualization in Excel

Data Analytics

Learning Objectives

PW SKILLS

By The End Of This Session, Participants Will Be Able To:

1. Create and structure Pivot Tables for effective data summarization.
2. Apply Value Field Settings and “Show Values As” techniques for advanced analysis (%, running totals, differences).
3. Develop Calculated Fields to derive meaningful business insights (e.g., Profit, Profit %).
4. Use Slicers and Timelines to add interactivity and dynamic filtering to reports.
5. Design and customize Pivot Charts for clear and impactful data visualization.
6. Build an interactive dashboard by integrating Pivot Tables, charts, slicers, and timelines.
7. Present data insights in a structured and visually appealing manner for decision-making.

Schedule Breakup

Section	Duration
Introduction to Pivot Tables & Use Cases	10 mins
Creating Pivot Tables (Quick Revision + Setup)	25 mins
Value Field Settings & Show Values As (% , Difference, Running Total)	25 mins
Calculated Fields (Profit, % Calculations)	20 mins
Break	10 mins
Slicers (Advanced Usage & Connecting Multiple Pivots)	25 mins
Timelines (Date-based Analysis)	20 mins
Pivot Charts (Design + Best Chart Selection)	20 mins
Building Interactive Dashboard (Layout + Integration)	15 mins
Wrap-up & Q&A	10 mins
Total	180 mins (3 hrs)

Introduction to Pivot Tables

1

- PivotTables help summarize large datasets dynamically.
- Enables filtering, sorting, and aggregating data efficiently.
- Fundamental for dashboard creation.

2

Why Pivot Table?

- Saves time.
- No complex formulas.
- Flexible grouping & filtering.

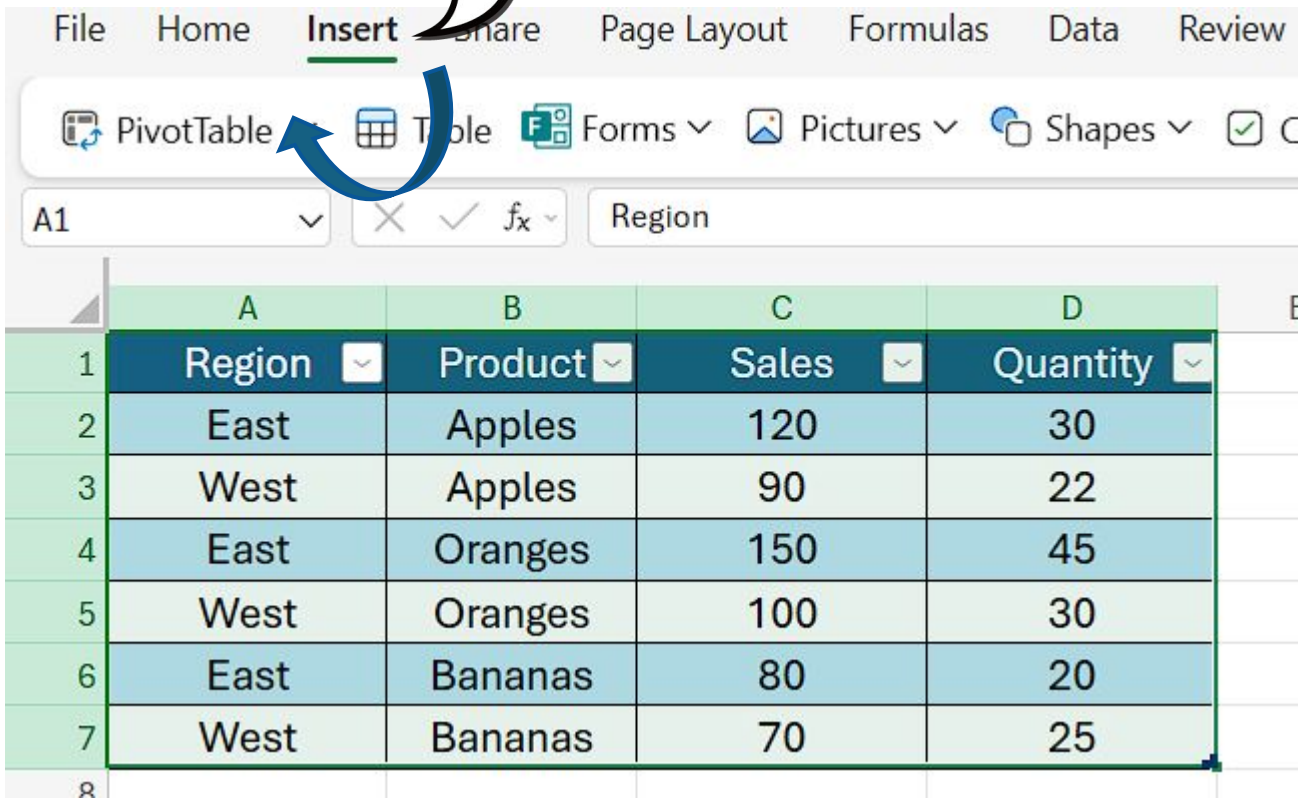


Step-By-Step Guide

1. Select the dataset
2. Go to Insert > PivotTable
3. Choose the range and destination
4. Drag fields into rows, columns, values, and filters

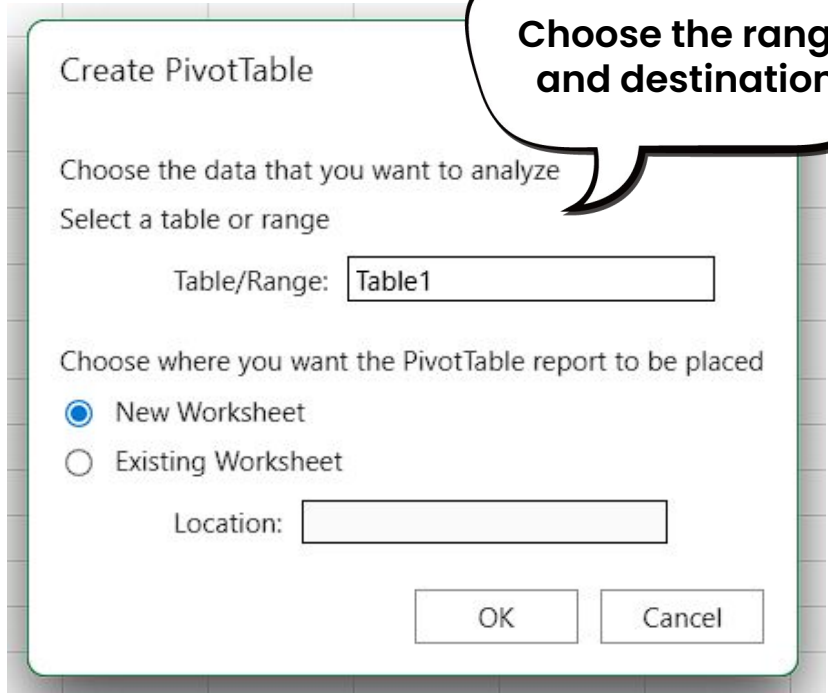


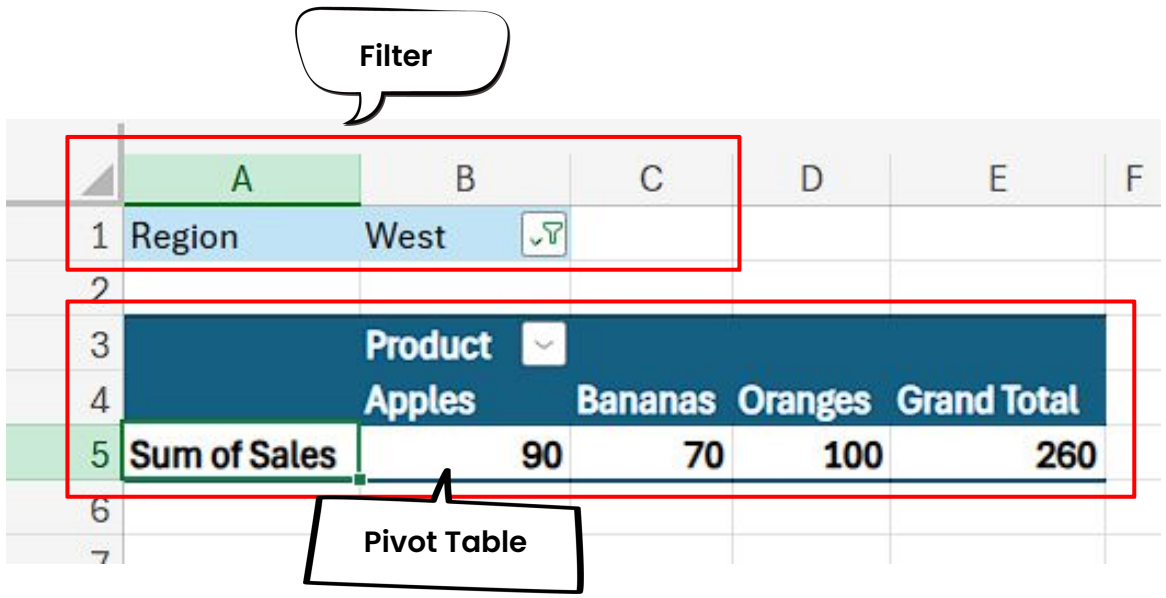
Select the dataset
Go to Insert > PivotTable

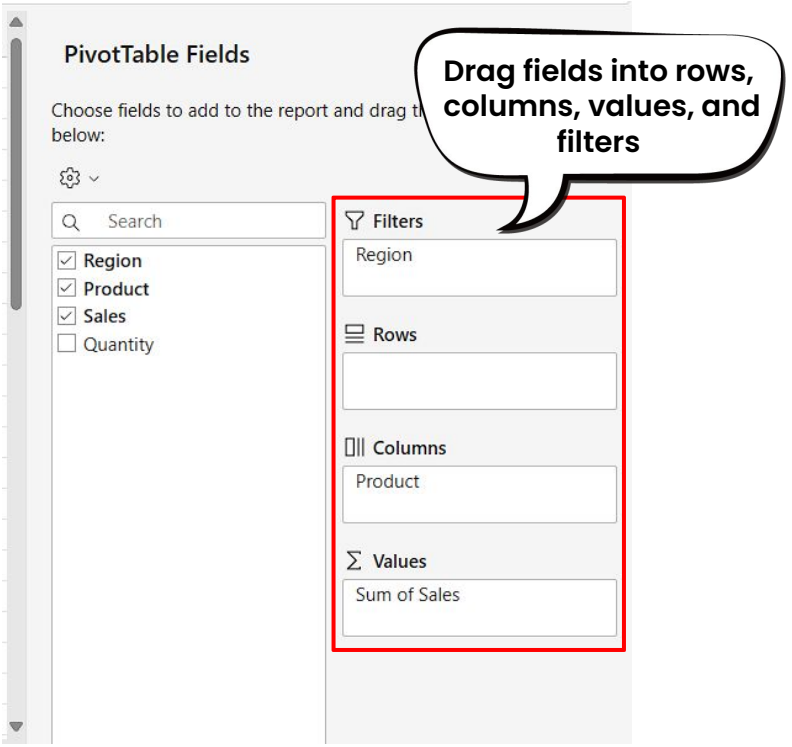


	A	B	C	D
1	Region	Product	Sales	Quantity
2	East	Apples	120	30
3	West	Apples	90	22
4	East	Oranges	150	45
5	West	Oranges	100	30
6	East	Bananas	80	20
7	West	Bananas	70	25

Choose the range and destination




Filter
A
B
C
D
E
F
1 Region West
2
3 Product
4 Apples Bananas Oranges Grand Total
5 Sum of Sales 90 70 100 260
6
Pivot Table


PivotTable Fields
Drag fields into rows, columns, values, and filters
Choose fields to add to the report and drag t
below:
Search
Region
Product
Sales
Quantity
Filters
Region
Rows
Columns
Product
Values
Sum of Sales

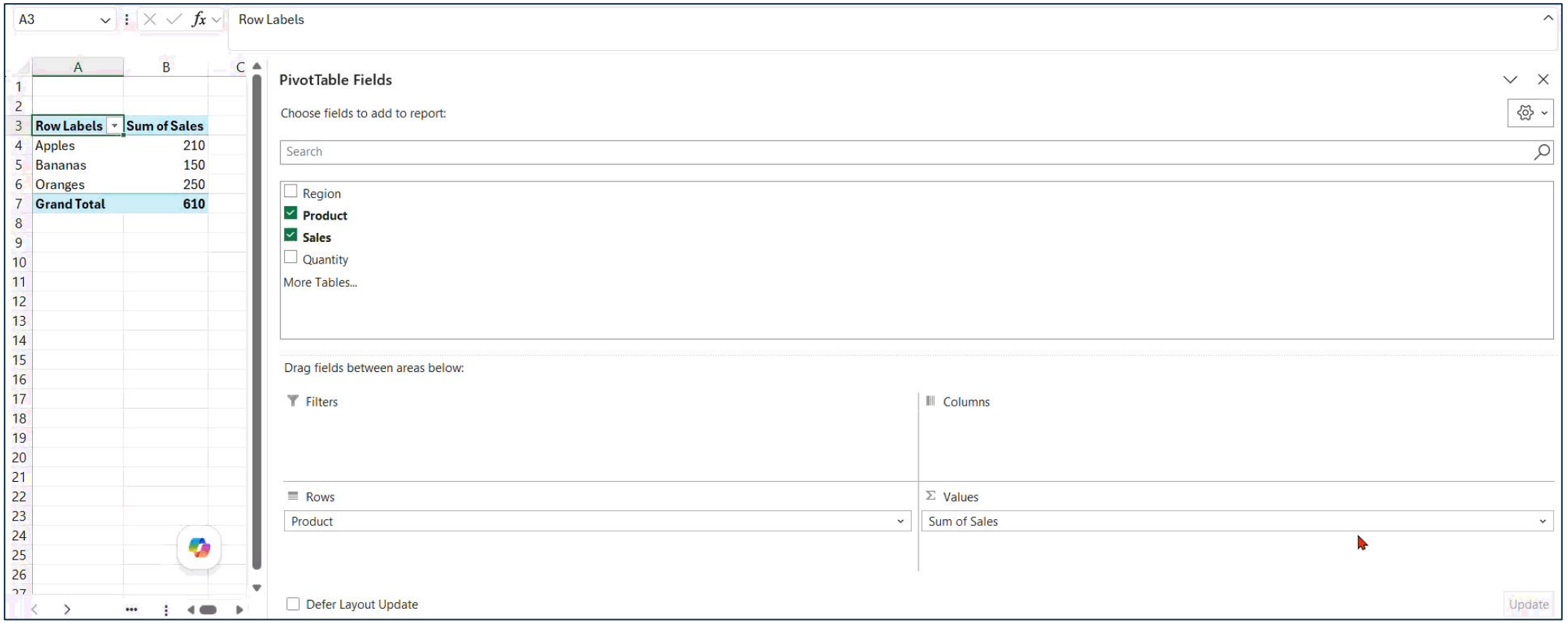
Value Field Settings & Show Values As

1. Controls how values are calculated in Pivot Tables

2. Default is Sum, but can be changed to:
   - Count
   - Average
   - Max / Min

3. Access: Right-click on value → Value Field Settings

Demo: Value Field Setting



Show Values As(Advanced Analysis)

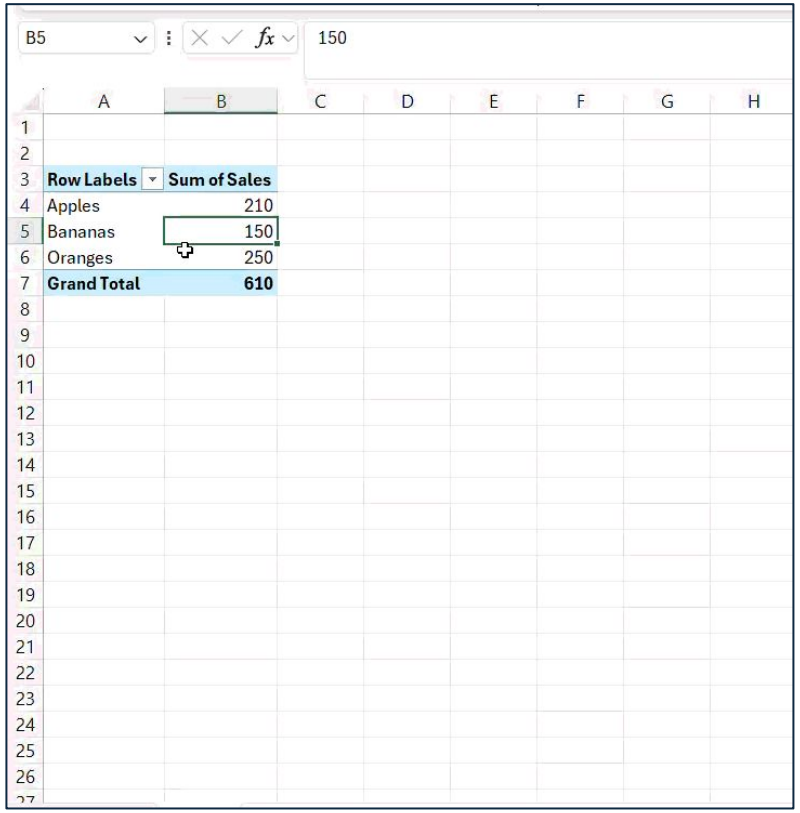
1

- Changes how values are displayed (not the actual data)
- Helps in comparisons and insights

2

Key Options

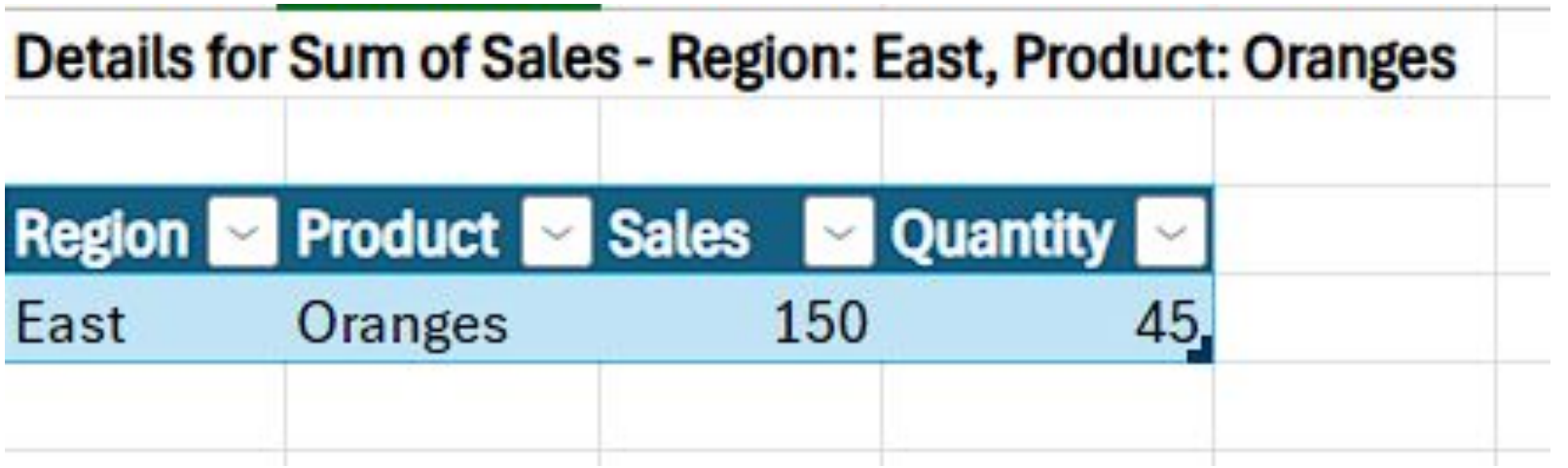
- % of Grand Total → Shows contribution to total
- % of Row/Column Total → Relative comparison
- Difference From → Compare with previous item
- Running Total In → Cumulative values



Drill-Down & Grouping in PivotTables

Drill-Down (Explore Details)

- Double-click on a number in PivotTable → opens the detailed rows from the dataset.
- Double-click on East → Oranges (150).



Groupin g

- If you had dates → group by Month/Quarter/Year.
- If you had larger Sales values → group into ranges (0–100, 101–200, etc.).
- Example: Group Products “Apples” + “Oranges” = “Fruits.”

PivotTable: Sales by Fruits vs Other

	Fruits	Other
East	270	80
West	190	70

Calculated Fields (Profit & % Calculations)

PW SKILLS

What is a Calculated Field?

- A custom formula created inside a Pivot Table
- Used to perform calculations using existing fields
- No need to modify the original dataset

Why Use Calculated Fields?

- Create new metrics (e.g., Profit, Profit %)
- Perform quick business analysis
- Avoid adding extra columns in source data

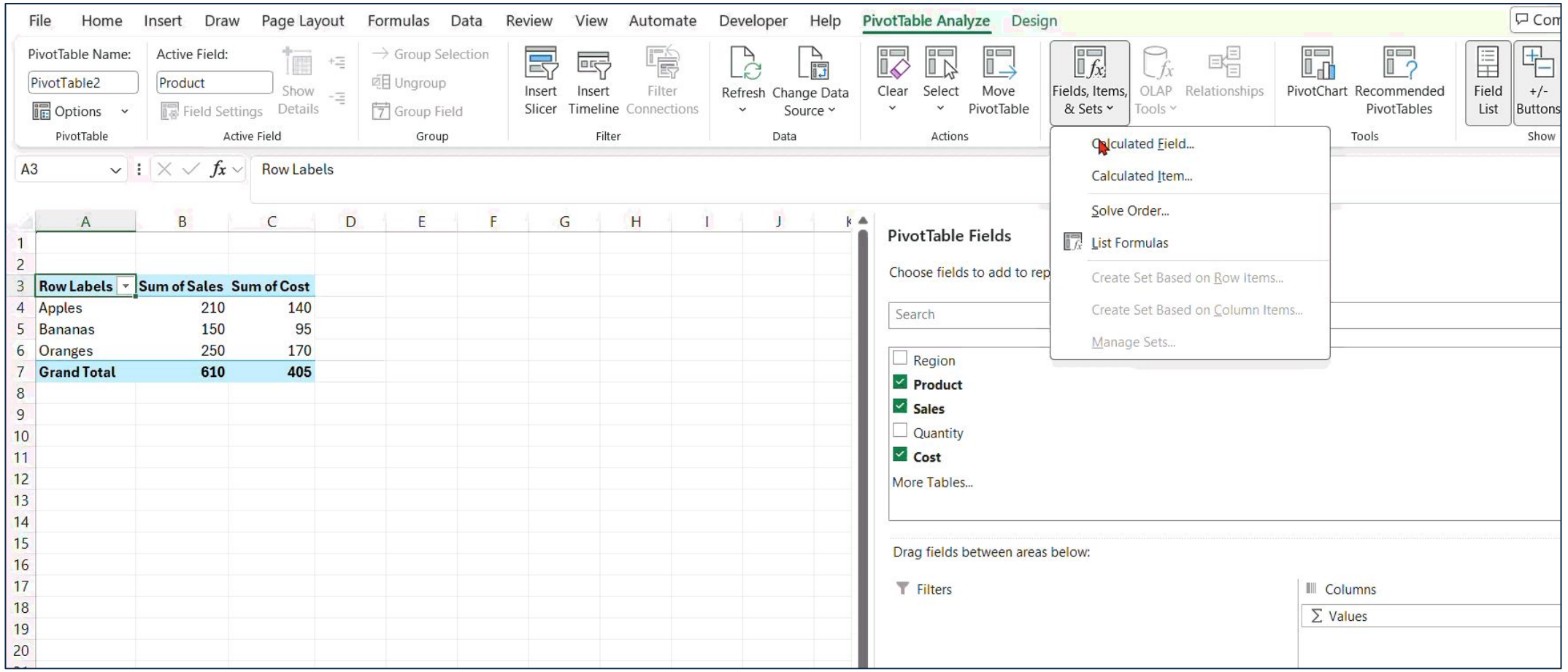
Example

Dataset Fields: Sales, Cost

- Profit = Sales – Cost
- Profit % = Profit / Sales

Steps to Create Calculated Field

1. Click anywhere inside the Pivot Table
2. Go to PivotTable Analyze tab
3. Click on Fields, Items & Sets
4. Select Calculated Field
5. Enter:
   - Name → Profit
   - Formula → = Sales – Cost
6. Click Add → OK


File Home Insert Draw Page Layout Formulas Data Review View Automate Developer Help PivotTable Analyze Design
PivotTable Name:
PivotTable2
Options
PivotTable
Active Field:
Product
Field Settings
Show Details
Active Field
Group Selection
Ungroup
Group Field
Group
Insert Slicer
Insert Timeline
Filter Connections
Filter
Refresh
Change Data Source
Data
Clear
Select
Move PivotTable
Actions
Fields, Items, & Sets
OLAP Tools
Relationships
PivotChart
Recommended PivotTables
Tools
Field List
+/- Buttons
Show
A3
Row Labels
Calculated Field...
Calculated Item...
Solve Order...
List Formulas
Create Set Based on Row Items...
Create Set Based on Column Items...
Manage Sets...
Row Labels
Sum of Sales
Sum of Cost
Apples 210 140
Bananas 150 95
Oranges 250 170
Grand Total 610 405
PivotTable Fields
Choose fields to add to rep
Search
Region
Product
Sales
Quantity
Cost
More Tables...
Drag fields between areas below:
Filters
Columns
Values

Percentage Calculation

Steps for Percentage Calculation

1. Go to Calculated Field again
2. **Enter:**
   - Name → Profit %
   - Formula → = (Sales - Cost) / Sales
3. Click Add → OK
4. **Format as Percentage (%)**

Row Labels	Sum of Sales	Sum of Cost	Sum of Profit%
Apples	210	140	0.33
Bananas	150	95	0.37
Oranges	250	170	0.32
Grand Total	**610**	**405**	**0.336065574**

“We’ve summarized data using Pivot Tables… now let’s make it interactive.”

“Introducing Slicers — for quick, visual, and user-friendly filtering.”

What is a Slicer?

- Slicers are interactive buttons that allow users to filter data visually.

- They improve user experience by making dashboards more intuitive.

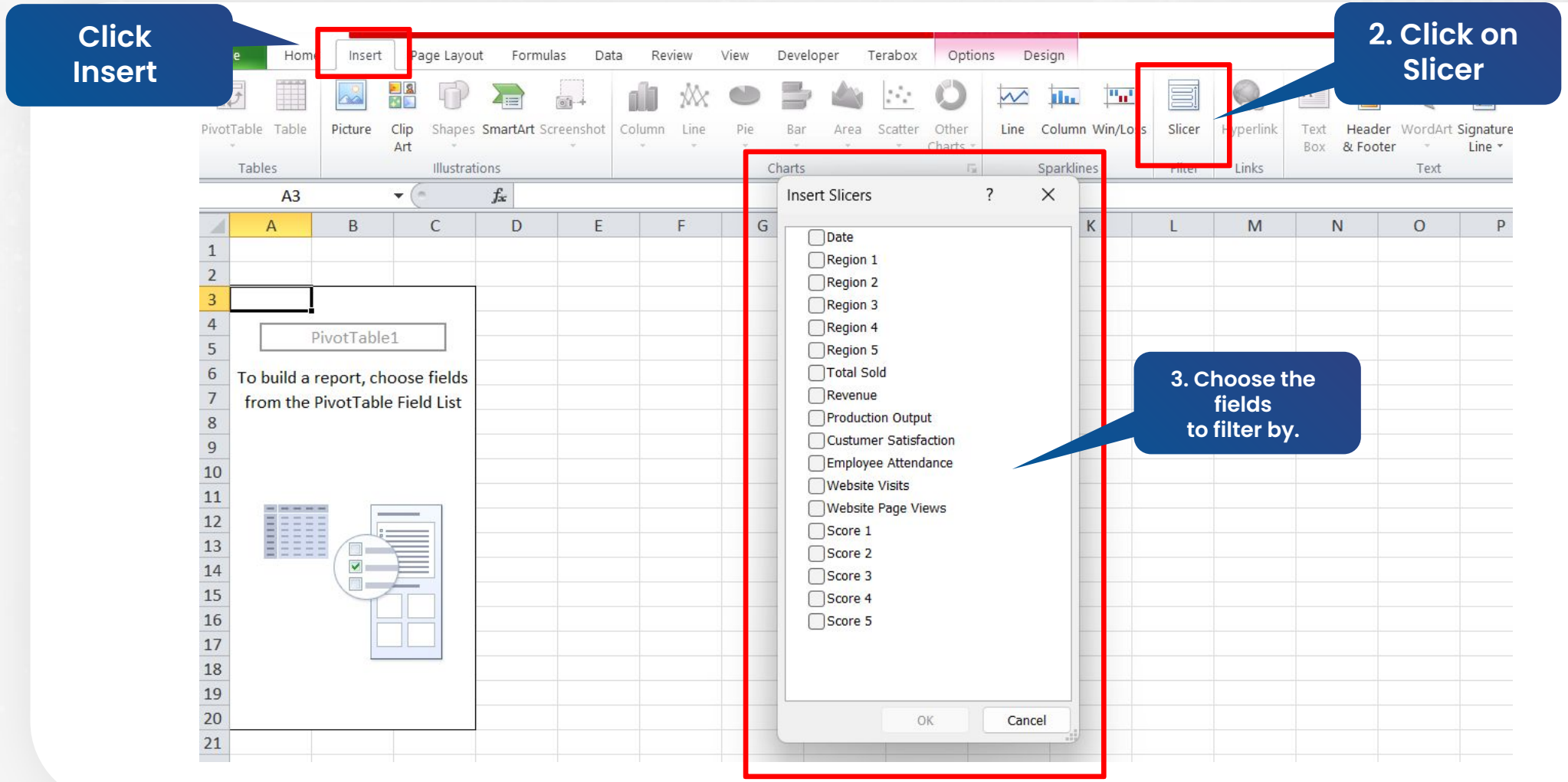
Key Features

- One-click filtering (no dropdowns needed)
- Easy to understand and use
- Can connect to multiple Pivot Tables
- Shows current filter selection clearly

Why Use Slicers?

- Makes dashboards interactive
- Saves time while analyzing data
- Improves data visualization & storytelling

Step-by-Step Guide



Using Timelines for Date Filtering

Timelines Allow Users To Filter Data Based On Dates Dynamically.

They are useful for financial and time-sensitive analysis.

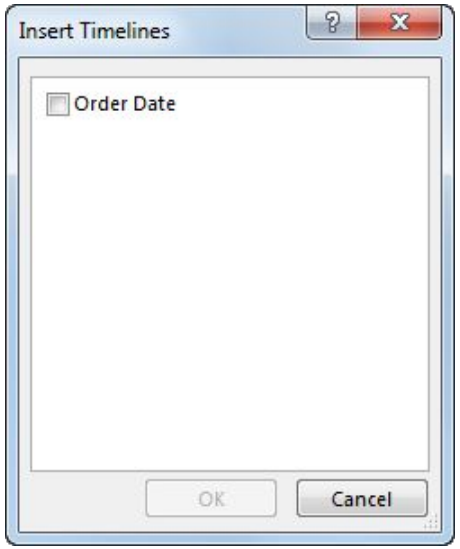
Key Features

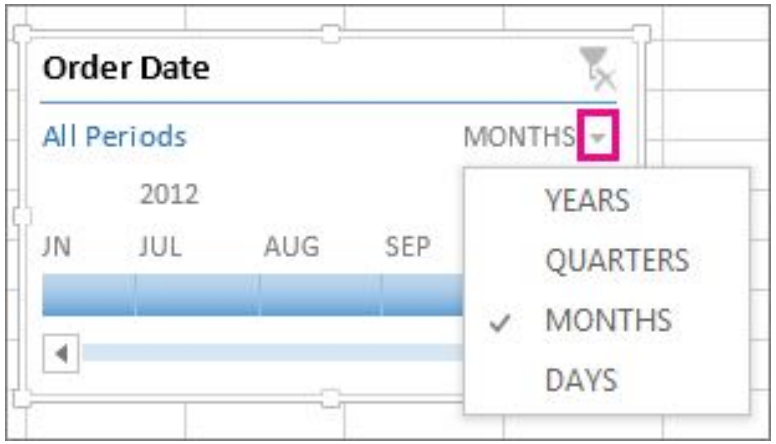
- Filters data by Year, Quarter, Month, or Day
- Easy drag-and-select timeline bar
- Works only with date-based data
- Shows time-based trends clearly

Step-by-Step Guide

- Click anywhere in a Pivot Table to show the PivotTable Tools ribbon group, then click Analyze > Insert Timeline.
- In the Insert Timeline dialog box, check the date fields you want, and click OK.
- Use the dropdown in the slicer to switch between Years, Quarters, Months, or Days.







Customize a Timeline

- Click on the timeline slicer.

- Use the Slicer Tools - Options tab to change the style, size, and appearance.



Timeline Styles

Why Use Timeline?

- Simplifies date filtering
- Makes analysis time-focused & interactive
- Ideal for sales trends, performance tracking

Slicer vs Timeline

Feature	Slicer	Timeline
Purpose	Filters any type of data	Filters only date fields
Data Type	Text, numbers, categories	Dates only
View Style	Buttons (click to filter)	Time-based slider bar
Ease of Use	Simple & flexible	Best for time analysis
Filtering	One-click selection	Drag to select time range
Best Use Case	Product, Region, Category	Sales trends, monthly/yearly analysis

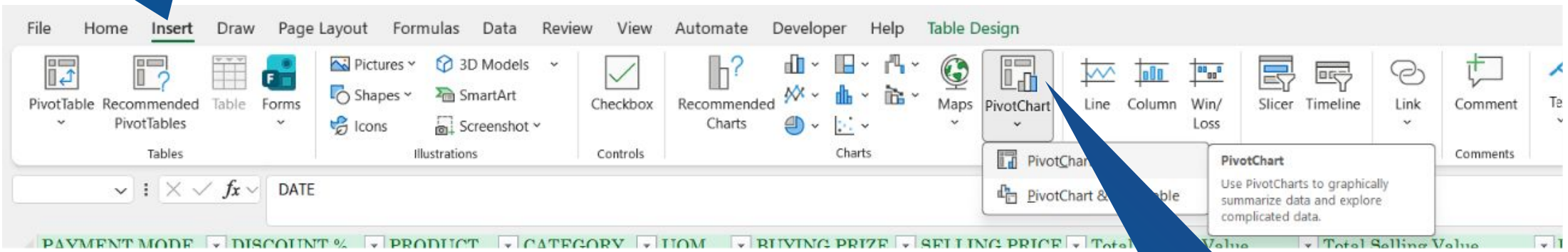
Creating PivotCharts

- PivotCharts visually represent Pivot Table data.

- They Provide Real-time Updates When Pivot tables Change.

Step-by-Step Guide

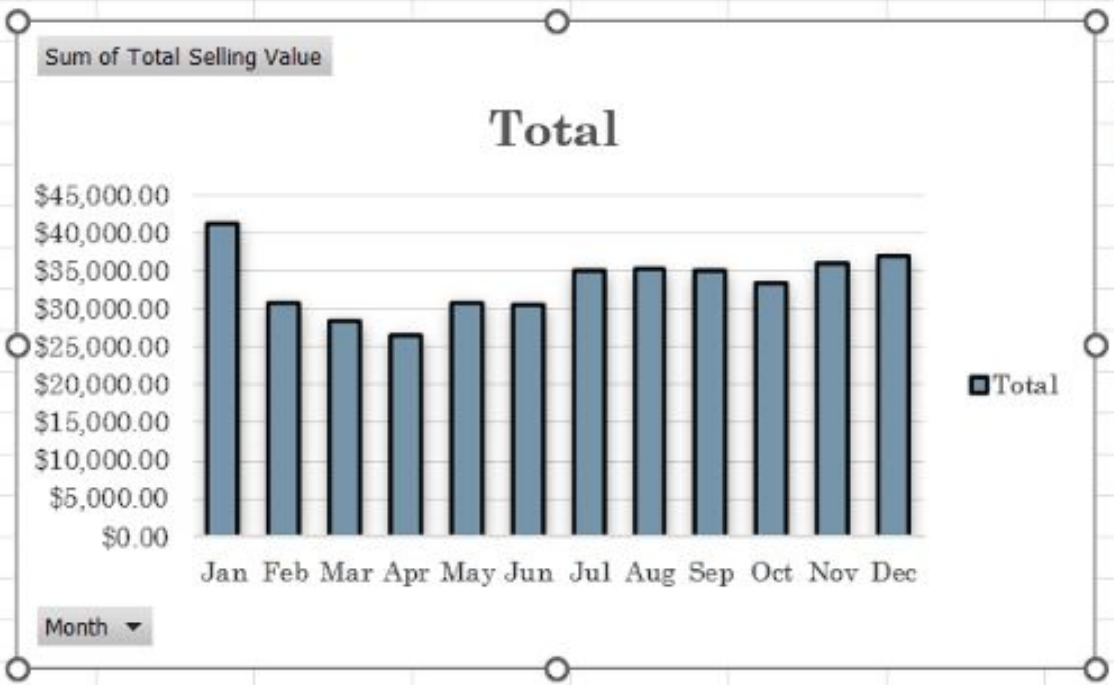
Click Insert



2. Select PivotChart

Example of PivotChart

Row Labels	Sum of Total Selling Value
Jan	$41,346.96
Feb	$30,857.30
Mar	$28,616.65
Apr	$26,579.11
May	$30,910.45
Jun	$30,533.71
Jul	$35,251.79
Aug	$35,350.40
Sep	$35,242.81
Oct	$33,500.69
Nov	$36,124.07
Dec	$37,097.98
Grand Total	**$4,01,411.92**

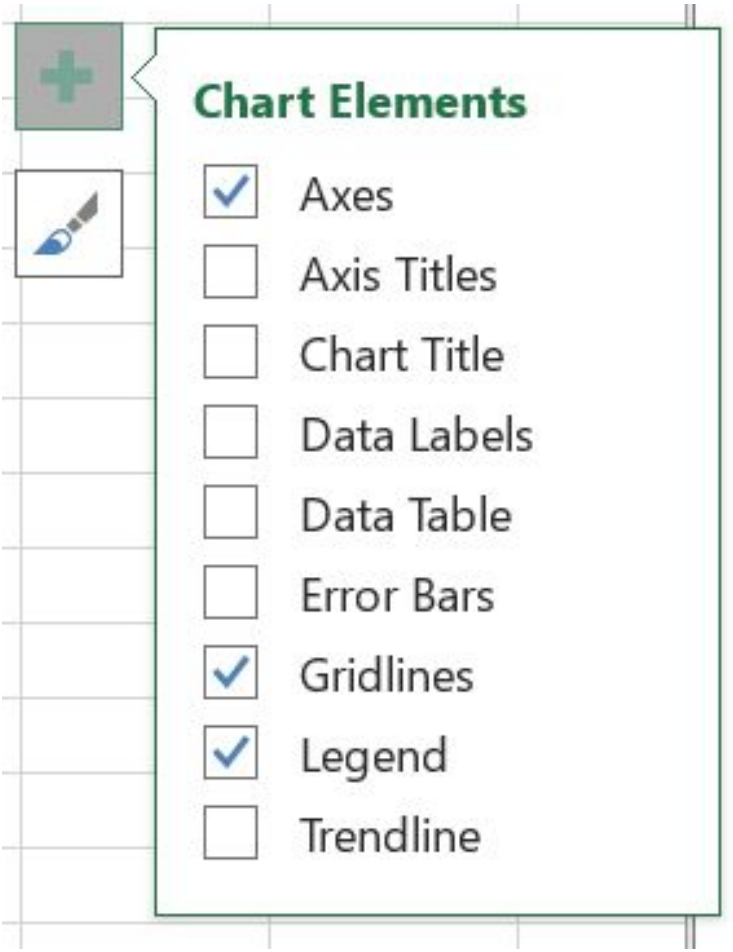


What are Chart Elements?

Options that help you customize and enhance your chart's appearance and readability.

Common Chart Elements

- **Axes** – Shows horizontal & vertical scale
- **Axis Titles** – Describes what each axis represents
- **Chart Title** – Main heading of the chart
- **Data Labels** – Displays exact values on chart
- **Gridlines** – Helps in easy data comparison
- **Legend** – Identifies different data series
- **Trendline** – Shows overall data trend



Pivot Charts & When to Use Them

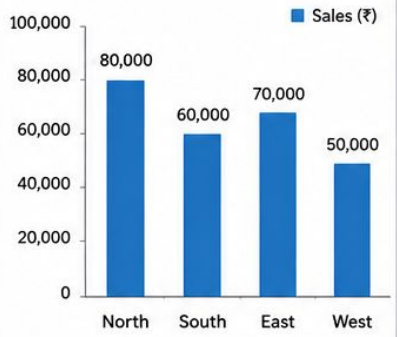
Different Charts for Different Insights

1. COLUMN CHART

Used for Comparison

Example: Sales by Region

Sales by Region



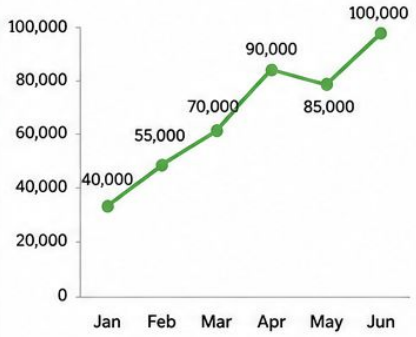
Best for comparing values across different categories.

2. LINE CHART

Used for Trends Over Time

Example: Monthly Sales Trend

Monthly Sales Trend



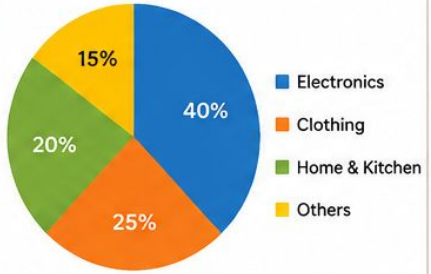
Best for showing trends and patterns over time.

3. PIE CHART

Used for Proportion / Share

Example: Sales by Category

Sales by Category



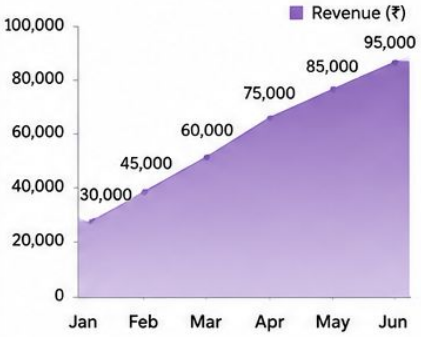
Best for showing percentage or contribution of each part.

4. AREA CHART

Used for Trend with Volume

Example: Revenue Over Time

Revenue Over Time



Best for showing trends along with magnitude (volume).

QUICK GUIDE – WHEN TO USE WHICH CHART?

- Compare Categories → **Column / Bar Chart**
- Show Trends Over Time → **Line Chart**
- Show Percentage / Share → **Pie Chart**
- Show Trend + Volume → **Area Chart**

KEY TAKEAWAY

Choose the right chart based on your data and the story you want to tell.

Pivot Chart vs Normal Chart

Feature	Pivot Chart	Normal Chart
Data Source	Based On Pivot Table	Based on regular data range
Interactivity	Highly Interactive (Filters, Slicers)	Limited interactivity
Updates	Auto-updates With Pivot Table Changes	Needs manual update
Flexibility	Easy To Rearrange Fields Dynamically	Fixed structure
Best Use Case	Dashboards & Analysis	Simple data visualization
Complex Data Handling	Handles Large Datasets Easily	Less efficient for complex data

Building a Complete Interactive Dashboard

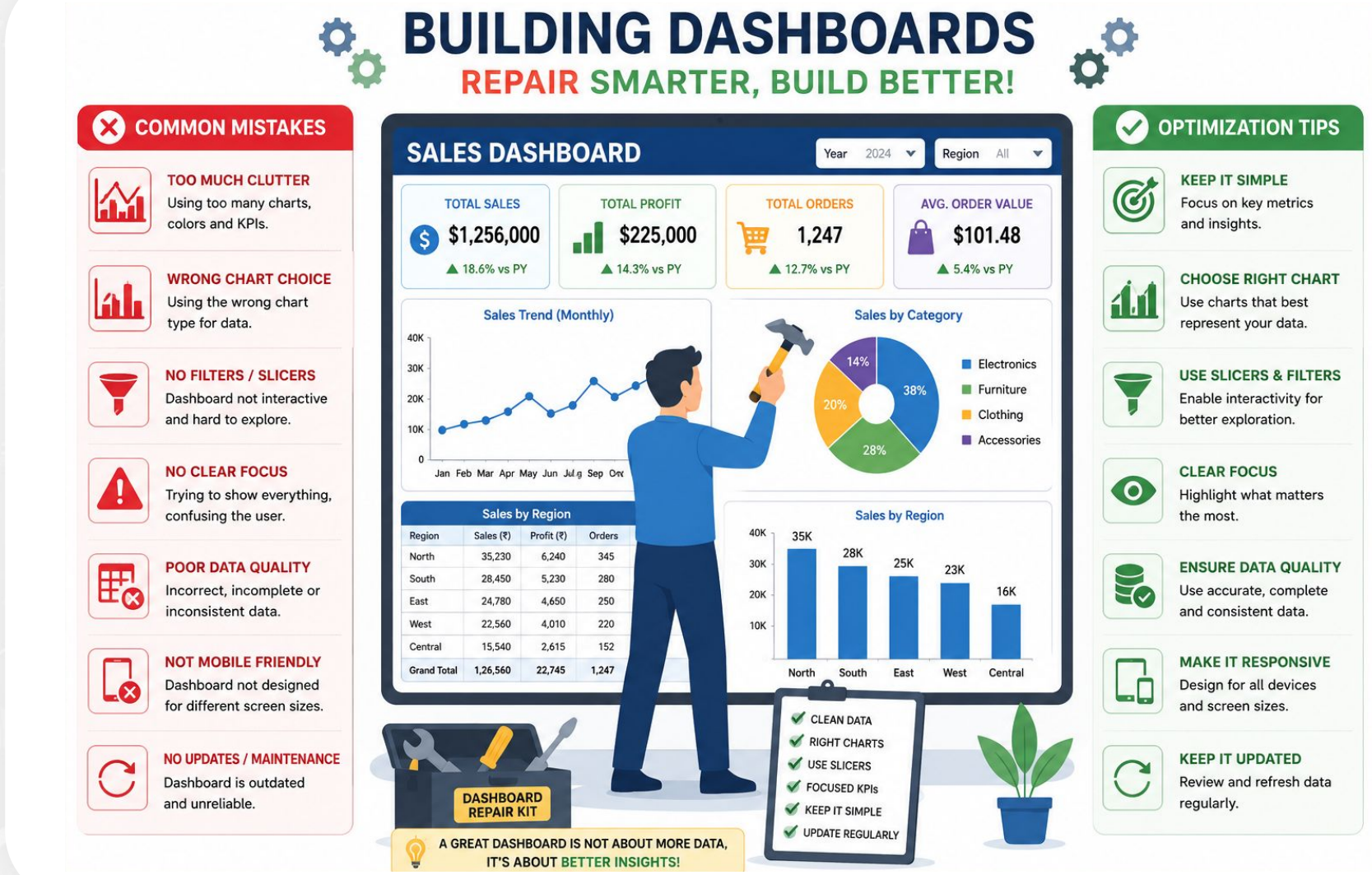
- A dashboard integrates multiple components for a complete view of data.
- Effective layout and formatting enhance user experience.

Step-by-Step Guide

- Organize data and create PivotTables.
- Add slicers and timelines for interactivity.
- Insert Pivot Charts for visualization.
- Arrange elements cohesively.


TITLE
Gives a clear description of the dashboard.
SALES DASHBOARD
Overview of Sales Performance
Year
2022
2023
2024
SLICER
Used to filter data interactively. (Here: Year, Region, Category)
KPI CARDS
Show key metrics at a glance for quick insights.
Total Sales
$1,26,560
▲ 18.6% vs PY
Total Profit
$22,745
▲ 14.3% vs PY
Total Orders
1,247
▲ 12.7% vs PY
Avg. Order Value
$101.48
▲ 5.4% vs PY
Region
North
South
East
West
Central
PIVOT CHART
Visual representation of pivot table data for better insights.
Sales by Month (Trend)
40K
30K
20K
10K
0
8K
9K
10K
12K
14K
11K
13K
15K
18K
20K
22K
25K
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Sales by Category
14%
38%
20%
28%
Electronics
Furniture
Clothing
Accessories
SLICER
Allows quick filtering by Region and Category.
Category
Electronics
Furniture
Clothing
Accessories
TABLE
Shows detailed values used as the data source for charts.
Sales by Region
Region | Sales (₹) | Profit (₹) | Orders | Avg. Order Value (₹)
North | 35,230 | 6,240 | 345 | 102.12
South | 28,450 | 5,230 | 280 | 101.61
East | 24,780 | 4,650 | 250 | 99.12
West | 22,560 | 4,010 | 220 | 102.55
Central | 15,540 | 2,615 | 152 | 102.24
Grand Total | 1,26,560 | 22,745 | 1,247 | 101.48
Sales by Region
40K
30K
20K
10K
0
35K
28K
25K
23K
16K
North
South
East
West
Central
PIVOT CHART
Compares values across categories (Here: Sales by Region).
This dashboard gives a quick overview of sales performance with the help of KPIs, Pivot Charts, Slicers and Tables.

Common Mistakes & Optimization



Session Summary

What We Covered

- Basics & **use cases of Pivot Tables**
- Creating & structuring Pivot Tables
- Advanced analysis using **Value Field Settings & Calculated Fields**

Interactivity & Visualization

- **Slicers & Timelines** for dynamic filtering
- Designing effective **Pivot Charts**
- Choosing the **right chart for the right data**

What is a Calculated Field?

- Combining **Pivot Tables + Charts + Slicers**
- Creating a **fully interactive dashboard layout**
- Best practices for **clear and impactful reporting**

PW SKILLS

Thank You

For Attending The Session!!

“Transform data into insights, and insights into decisions. ”

Keep Learning, Keep Upskilling!